Assessment 3: Project Specifications

Due Date: Multiple Deadlines (see submission pages)
Points: None
Weighting: 40%
Group Size: Min 3, Max 3
Type: Project (Group & Individually assessed)
Reference: Week 1 Potential Topics

Overview

Purpose: To ensure that students have a clear understanding of the theoretical concepts, are able to implement them in a real-world computer vision problem, and can independently/collaboratively design and implement solutions using these techniques.

Learning outcomes: SLO1, SLO2, SLO3, SLO4, SLO5

Important Administrative Notes:

- Please check individual submission pages for deadlines, respective deliverables, report templates, etc.
- Assignment-3 is incomplete without the presentation, demo, and viva/oral defense.
- Marks for intermediate deliverables will be awarded only after the project has been completed in full.
- Final marks will be allocated proportionally based on each member's individual contribution to the project, based on the agreed contribution in the final report.

Deliverables

Multiple Deliverables:

- **Part A (5 pts):** Project title, Abstract, Group details
- **Part B (5 pts):** Implementation/Development Plan
- **Part C (5 pts):** Initial Experimental Results
- **Part D (5 pts):** GUI Design
- **Part E (40 pts):** Complete Project Report

- **Part F (15 pts):** In-person presentation & Demo
- **Part G (25 pts):** Oral Defense

None required on this page. For information only.

Task Requirements

- Must be a Computer Vision or a combination Computer Vision and techniques (NLP, etc.).
- Develop a clear problem statement that is within the capabilities of CNNs, design and capture a dataset of significant size that addresses a problem in Computer Vision.
- Design/Implement an image classification algorithm, or
- Design/develop an object detection system for detecting specific objects in a video or live stream and localizing them, to solve a business/real-world problem.
- Compare a series of algorithms against each other to determine optimum performance, and then suggest new approaches that improve performance.

GUI Development:

Depending on the project topic, an exclusive interface (Graphical User Interface) must be developed, appropriate to operationalize the DL solution.

Options: Web App, Mobile App, Desktop App, Edge Device, etc.

Expected Evidence:

- Multiple deliverables as outlined above.
- Google Colab/AWS Sagemaker or IPython notebook, with the code or code repository. Code submission link will be provided by the subject coordinator after presentation and oral defense.
- It is expected that all team members must contribute equally in the project development and delivery.
- Oral defense, presentation, recorded video (see individual submission page for details).

Use of AI in this Assessment

You are permitted to use GenAI apps for the following purposes **only**:

- GenAI **must not** be used to generate code and report.
- GenAI can be used for feedback on your drafts, polish writing.
- GenAI can be used as mentioned above, as long as you can explain and defend your work when required.

Declaration Rules:

1. Declare usage using APA citation format: UTS Library guidance.
2. Save your chat transcript with the app and upload the file with your submission, declaring at the top that this is a true transcript.
3. Provide a 100-words written justification for using the GenAI app and reflect on whether you found it useful or not, with screenshot examples to evidence your points (see assessment grading rubric for details).

Administrative & Support

Feedback and support:

- Students will receive feedback during tutorial sessions or weekly consultation sessions.
- A Project Dashboard will be released after Part-A submission, feedback provided to each team will be recorded on the project Dashboard.
- Students to get early feedback on their assessment drafts, either from the subject coordinator during the weekly consultation session or speak to tutors as well.
- Students can also check the Project Dashboard for specific feedback on their projects. For questions related to this assignment, attend the Weekly Consultation Session to discuss with the Subject Coordinator or post your question on the discussion board.

Academic integrity: Academic integrity is about demonstrating honesty, trust, fairness, respect, responsibility and courage in your studies. A breach of academic integrity is known as 'academic misconduct' and happens if you engage in behaviours that undermine academic integrity, such as plagiarism and cheating. These are serious forms of misconduct and penalties apply. If you're thinking about using GenAI in your subjects and assessments, check the rules for whether and how you can use it for this subject via your subject site, tutor, lecturer or subject coordinator. You should also understand how to use GenAI ethically and reference and acknowledge its use.

IMPORTANT NOTE

Your assignment is INCOMPLETE without in-person presentation, oral defense.